

Birla Institute of Technology and Science, Pilani

Quantum Machine Learning (AIMLZG545)

Assignment 1: Literature Survey Report

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Objective: Identify a classical problem with scaling limitations.

Problem Chosen: Molecular Ground State Energy Estimation

(a) What is the Problem?

The **Molecular Ground State Energy Estimation** problem involves computing the lowest energy state (ground state) of a molecular system. The ground state energy is the minimum eigenvalue of the molecular Hamiltonian $\hat{H}$, obtained by solving the Schrödinger equation:

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

The ground state energy E_0 determines how a molecule behaves — its stability, reactivity, and interactions with other molecules.

Why This Problem Matters — Drug Discovery Example

The most compelling application is **drug discovery**. To design a new medicine, scientists must predict how a candidate drug molecule interacts with a target protein in the body. This requires computing the ground state energy of the drug-protein molecular complex.

- **Binding affinity:** Whether a drug binds to a protein depends on the ground state energy of the combined system. Lower energy = stronger binding = more effective drug.
- **Molecular screening:** Pharmaceutical companies need to screen millions of candidate molecules. Each screening requires a ground state energy calculation to predict if the molecule will work as a drug.
- **Side effects:** Predicting unwanted drug-protein interactions (side effects) also requires computing ground state energies of the drug with off-target proteins.
- **The bottleneck:** A typical drug-protein system involves hundreds to thousands of electrons. Computing its ground state energy exactly is classically impossible due to the exponential scaling described below.

Today, drug development takes 10–15 years and costs \$2–3 billion per approved drug, largely because classical computers cannot accurately simulate molecular interactions at the quantum level.

The Scaling Problem

The molecular Hamiltonian $\hat{H}$ for n electrons is a matrix of size $2^n \times 2^n$. As the number of electrons increases, this matrix grows *exponentially*, making it fundamentally intractable for classical computers beyond small molecules.

Molecule	Electrons	State Space Size	Classical Feasibility
Water (H ₂ O)	10	$2^{10} = 1,024$	Tractable
Iron-Sulfur cluster	30	$2^{30} \approx 10^9$	Difficult
Caffeine (C ₈ H ₁₀ N ₄ O ₂)	102	$2^{102} \approx 10^{30}$	Impossible
Protein-drug complex	~ 1000 s	2^{1000}	Completely impossible

(b) Why is it Difficult to Solve Classically?

The difficulty arises from a fundamental mismatch: **nature is quantum, but classical computers are not.**

Root Cause: Exponential State Space

A system of n quantum particles (electrons) can exist in a superposition of 2^n basis states simultaneously. To describe this system, a classical computer must store all 2^n complex amplitudes explicitly in memory. This leads to:

- **Exponential memory requirement:**
 - 20 electrons $\rightarrow 2^{20} \approx 10^6$ parameters (feasible)
 - 40 electrons $\rightarrow 2^{40} \approx 10^{12}$ parameters (terabytes of RAM)
 - 50 electrons $\rightarrow 2^{50} \approx 10^{15}$ parameters (~ 8 petabytes — exceeds all RAM on Earth)
- **Exponential time complexity:** The exact classical method (Full Configuration Interaction, FCI) requires diagonalizing the $2^n \times 2^n$ Hamiltonian matrix, with time complexity $O(2^n)$. Even on the fastest supercomputer, this is infeasible for $n > 20$.
- **Electron correlation:** Electrons interact via Coulomb repulsion. Each electron's state depends on ALL other electrons simultaneously, preventing any simplification by treating electrons independently.
- **No efficient classical algorithm exists:** Unlike problems such as sorting or graph search, where clever algorithms reduce complexity, the exponential scaling here is intrinsic to the physics — classical bits cannot efficiently represent quantum states.

Approximate Classical Methods (All Have Trade-offs)

Method	Scaling	Max Size	Accurate?	Key Limitation
Hartree-Fock	$O(M^4)$	Large	No	Ignores electron correlation
DFT	$O(M^3)$	Large	Sometimes	Fails for strongly correlated systems
CCSD(T)	$O(M^7)$	~ 50 atoms	Usually	Too expensive for large molecules
FCI (Exact)	$O(2^n)$	~ 20 electrons	Yes	Exponentially expensive

Brief overview of each method:

- **Hartree-Fock (HF):** Treats each electron as moving in the average field of all other electrons. It ignores the instantaneous electron-electron repulsion (correlation), so it gives qualitatively correct but quantitatively inaccurate results. Fast ($O(M^4)$) but errors can be 5–10%.
- **Density Functional Theory (DFT):** Instead of tracking each electron’s wavefunction, it uses the total electron density $\rho(\mathbf{r})$ to compute energy. Scales as $O(M^3)$ and works well for many systems, but relies on approximate “exchange-correlation functionals” that fail for strongly correlated molecules (e.g., transition metal catalysts).
- **CCSD(T) — Coupled Cluster:** Systematically accounts for electron pairs (doubles) and triples of interacting electrons. Often called the “gold standard” of quantum chemistry due to high accuracy. However, its $O(M^7)$ scaling limits it to ~ 50 atoms — far too small for drug-protein systems.
- **FCI — Full Configuration Interaction:** The only exact classical method. It considers *every possible* arrangement of electrons. Guarantees the correct answer but scales as $O(2^n)$ — making it usable only for molecules with ≤ 20 electrons (≤ 10 atoms).

No classical method exists that is simultaneously fast, accurate, and general-purpose.

(c) What are the Key Limitations?

The key scaling limitations that make this problem classically intractable are:

1. **Time:** The exact method (FCI) scales as $O(2^n)$. For a caffeine molecule with 102 electrons, the computation would require more time than the age of the universe ($\sim 10^{10}$ years), even on the world’s fastest supercomputer.
2. **Dimensionality (Curse of Dimensionality):** The quantum wavefunction $|\psi\rangle$ lives in a 2^n -dimensional Hilbert space. Each additional electron *doubles* the number of parameters needed. This exponential growth cannot be avoided without sacrificing accuracy.
3. **Memory/Data:** Storing the full wavefunction of just 50 electrons requires ≈ 8 petabytes of RAM — more than any existing computer possesses. Even representing the problem input exceeds classical hardware limits.
4. **Accuracy vs. Speed Trade-off:**
 - Fast approximate methods (DFT, Hartree-Fock): errors of 1–5%, unreliable for important systems.
 - Accurate methods (FCI, CCSD(T)): exponentially slow or $O(M^7)$, limited to small molecules.

No classical algorithm achieves both chemical accuracy (< 1 kcal/mol error) and polynomial scaling.

5. **Strong Correlation Problem:** Many industrially important molecules — catalysts (FeMoCo), high-temperature superconductors, biological enzymes — are “strongly correlated,” meaning all classical approximation methods break down completely for these systems.

Summary: The molecular ground state energy problem has *inherent exponential scaling* (2^n) in time, memory, and dimensionality. This is not a limitation of current algorithms but a fundamental mismatch between classical computation and quantum physics.

(d) References

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